

Speaking notes

Code changes: from the spin-dependent NNN model to the checkerboard model

1 Title

Record of every code change, old NNN model to checkerboard. Two codebases: Fortran and Python.

2 3 of 10 files

Only 3 files changed. Engine NEVER touched: libuph.f90, 2699 lines, byte-identical.

So any difference in results is physics, not the sampler.

3 The idea

OLD: the two spins hop differently, splitting built in at $U=0$.

NEW: both spins hop identically, the GEOMETRY alternates instead. Splitting must emerge from U .

4 mc2duph.f90 (KEY)

The whole physics change is one if-statement.

Before: $tk(\dots, 1)$ and $tk(\dots, 2)$ carry OPPOSITE signs, same at every site.

After: same value for both spins; what alternates is parity $\text{mod}(\text{ixv}+\text{iyv}, 2)$. $tp=t1+t2$, $tm=t1-t2$.

5 Verification

Checked numerically, not asserted. $\max \text{abs}(K_{\text{up}} - K_{\text{dn}})$: 0.6 old $\rightarrow$ 0.0 new.

Splitting at fixed k : 2.4 old $\rightarrow$ 0.0 new. Old has d_{xy} form $-8*t1*\sin(kx)\sin(ky)$.

IF ASKED: sorted spectra give zero for BOTH models. The two spin Hamiltonians are related by a 90-deg rotation, so eigenvalue SETS match. That degeneracy IS the compensation -- and is exactly why we measure the k -resolved Delta $n(k)$ and not the total magnetization, which is zero in both.

6 parameter.f90

One line. Fortran needs every variable declared. Bookkeeping only, nothing else touched.

7 wf_unified.py GetK

The trial must describe the SAME model, or the constrained path is defined by a

Hamiltonian we are not solving. Signature $(t, tA, tt) \rightarrow (t, t\text{prime}, \text{delta})$. Checked element by element.

8 wf_unified.py neel_seed

Follows from removing the spin dependence. The new H is spin-symmetric, so a

symmetric start STAYS symmetric under Hartree-Fock. Seed 0.9 up on A, 0.9 down on B to break it.

9 Summary (Fortran)

Old model IMPOSES altermagnetism. New model lets it EMERGE. Everything downstream unchanged.

10 Python: opposite

Fortran 3 modified / 0 added. Python 0 modified / 13 added, 939 lines.

Why: CPMC(..., $K=\dots$) takes the hopping as an ARGUMENT, so a new lattice is only a new matrix.

cpqmc.py last changed by a commit that predates all checkerboard work.

11 checkerboard.py

Same parity rule in both languages: $\text{mod}(\text{ixv}+\text{iyv}, 2)$ vs $(x+y)\%2$, same tp and tm .

Not left to inspection -- Gate 1 requires $\max \text{abs}(\text{difference}) = 0$ vs the Fortran matrix.

12 diag_bond_factors (KEY)

New PHYSICS, not a port. Old code had s , extended s , $d_{x^2-y^2}$ -- all on

NEAREST-NEIGHBOUR bonds. $d_{xy} \sim \sin(kx)\sin(ky)$ lives on the DIAGONALS, so it had no form factor and could not be measured at all. Since d_{xy} pairing is our main result, the observable had to be added too.

13 Validation

Four ED gates before use. Gate 3 is the important one: the connected vertex must vanish at

$U=0$ by construction, so a nonzero value would mean the new form factor or the contraction is wrong.

14 Summary / next week

Emergent AM; d_{xy} at 126/126 cells across $L=8, 10, 12$.

Anisotropy ENHANCES d_{xy} and suppresses $d_{x^2-y^2}$ -- the earlier opposite claim came from $L=6$.

Magnetism and pairing pick the SAME channel. Long-range order NOT established: $S(\pi, \pi)/N$ falls as $1/N$, $h \rightarrow 0$ intercept falls with L . $L=16$ is running. NEXT WEEK: finish writing the manuscript.

LIKELY QUESTIONS

"Is the engine really unchanged?" $\rightarrow$ libuph.f90 is byte-identical, 2699 lines. Diff it live if he wants.

"How do you know the new model has no splitting?" $\rightarrow$ K_{up} and K_{dn} are bit-for-bit equal. Slide 5.

"Why did the d_{xy} conclusion flip?" $\rightarrow$ the old claim was $L=6$. $L=8, 10, 12$ all give the opposite, and the vertex GROWS with size (2.3-3.2x from $L=8$ to $L=12$). We no longer quote $L=6$ for anything.

"Is the altermagnetism real?" $\rightarrow$ splitting yes, long-range ORDER not established at these sizes. Say so.